

Ayush Choudhary

23f3000370

23f3000370@ds.study.iitm.ac.in

Diploma level

Description

This project is a **Flask-based Quiz Application** that allows **users** to attempt quizzes and **admins** to manage subjects, chapters, and quiz questions. It includes **user authentication**, **quiz attempts tracking**, and **score summaries** displayed using **charts**.

Technologies Used

- **Flask** – Web framework for handling routes and views
- **SQLite** – Lightweight database for storing quiz data
- **Flask-Session** – Managing user sessions
- **Jinja2** – Template engine for rendering dynamic HTML pages
- **Chart.js** – Visualization library for summary charts
- **Flask Flash** – Displays feedback messages to users (e.g., success/error notifications)

Purpose of Technologies:

- Flask simplifies backend development and routing.
- SQLite provides an embedded, easy-to-manage database.
- Chart.js helps visualize quiz statistics interactively.
- Flask Flash provides real-time feedback to users for actions like adding quizzes or handling errors.

DB Schema Design

Tables

1. **SUBJECTS** (*id*, *subject_name*)
 - Stores quiz subjects.
2. **CHAPTERS** (*id*, *chapter_name*, *subject_id*, *no_of_questions*)
 - *subject_id* as **Foreign Key** referencing **SUBJECTS**(*id*).
3. **QUIZES** (*id*, *quiz_name*, *chapter_id*, *date*, *duration*)
 - *chapter_id* as **Foreign Key** referencing **CHAPTERS**(*id*).
4. **QUESTIONS** (*id*, *quiz_id*, *question*, *option1*, *option2*, *option3*, *option4*, *answer*)
 - *quiz_id* as **Foreign Key** referencing **QUIZES**(*id*).
5. **USERS** (*id*, *username*, *password*, *full name*, *qualification*, *dob*)
 - Role: **user**
6. **QUIZ_ATTEMPTS** (*id*, *user_id*, *quiz_id*, *score*, *attempt_date*)
 - Tracks user quiz attempts.
7. **ADMIN** (*id*, *username*, *password*)
 - Predefined username and password for admin role

Schema Design Reasoning:

- **Foreign keys** ensure data consistency.
- **Separate tables** for modularity (subjects → chapters → quizzes → questions).

API Design

- **User Authentication:** `/login`, `/logout`, `/signup`, `/admin_login`
- **Subject Management:** `/add_subject`, `/edit_subject`, `/delete_subject`
- **Chapter Management:** `/add_chapter`, `/edit_chapter`, `/delete_chapter`
- **Summary:** `/admin_summary`, `/user_summary`
- **Quiz Management:** `/add_quiz`, `/edit_quiz`, `/delete_quiz/view_quiz`
- **Question Management:** `/add_question`, `/edit_question`, `/delete_question`
- **Quiz Attempts:** `/attempt_quiz`, `/submit_quiz`, `/view_results`
- **Dashboard Stats:** `/admin_dashboard`, `/user_dashboard/search_admin`

Each endpoint interacts with the database using **SQLite (sqlite3 library)**, processes POST/GET requests, and returns JSON responses or HTML templates.

Architecture and Features : Project Structure:

```
/Quiz_Master_V1
|— /templates      # HTML templates
|— /static         # CSS, JS, images
|— /controllers    # Routes and API logic
|— /models         # Database models
|— app.py          # Main application
|— quiz_master.db  # SQLite database
```

Implemented Features:

- ✓ **User Authentication** – Secure login & role-based access
- ✓ **Quiz Management** – CRUD operations for subjects, chapters, quizzes, and questions
- ✓ **Quiz Attempt & Scoring** – Users can take quizzes, scores are stored
- ✓ **Dashboard with Charts** – Admin/User dashboards with **Chart.js** visualizations

Presentation Video link:

https://drive.google.com/file/d/1BigbanJ16dltF-mYcx_GOGXX_7hp5LbI/view?usp=sharing

